

Product Requirements Document

Slime Soccer

Document Version	1.0
Last Updated	December 2024
Product Type	2D Physics-Based Sports Game
Platform	Web Browser (HTML5/JavaScript)

1. Executive Summary

Slime Soccer is a fast-paced, retro-styled 2D physics-based soccer game featuring two semicircular slime characters competing to score goals. The game combines accessible controls with satisfying physics mechanics, offering both single-player gameplay against an AI opponent and local multiplayer for competitive play between friends.

The core appeal lies in the bouncy, responsive feel of the slime characters combined with realistic ball physics that reward skillful play. With configurable match durations and an anti-camping system to maintain exciting gameplay, Slime Soccer delivers quick, engaging matches perfect for casual gaming sessions.

2. Product Vision & Goals

2.1 Vision Statement

Create an addictive, pick-up-and-play soccer experience that captures the joy of classic arcade sports games while leveraging modern physics simulation for satisfying, skill-based gameplay.

2.2 Primary Goals

1. Deliver intuitive controls that new players can learn in seconds but take time to master
2. Implement physics that feel responsive and rewarding, creating emergent gameplay moments
3. Provide challenging AI that adapts to be fun for players of varying skill levels
4. Support both solo and social play through single-player and local multiplayer modes
5. Maintain engaging match flow through anti-camping mechanics

3. Target Audience

3.1 Primary Users

- **Casual gamers** seeking quick, fun gaming sessions during breaks
- **Retro gaming enthusiasts** who appreciate classic arcade aesthetics
- **Friends/families** looking for local competitive games
- **Students/office workers** wanting browser-based entertainment

3.2 User Demographics

- Age range: 8-45 years
- Gaming experience: Beginner to intermediate
- Platform preference: Desktop/laptop web browsers

4. User Stories

1. **Quick Play:** As a casual player, I want to start a game quickly so I can enjoy a match during a short break.
2. **Match Duration:** As a player, I want to choose how long my match lasts so I can fit games into my available time.
3. **Skill Development:** As a player, I want responsive controls and consistent physics so I can improve my technique over time.
4. **Local Competition:** As a player with friends nearby, I want to play against them on the same computer for head-to-head competition.
5. **Fair AI Challenge:** As a solo player, I want an AI opponent that is challenging but beatable so I stay engaged without frustration.
6. **Active Gameplay:** As a player, I want the game to discourage camping so matches remain exciting and competitive.

5. Game Modes

Mode	Description
Single Player	Player (cyan slime, left side) vs AI opponent (red slime, right side). AI provides adaptive challenge suitable for practice and solo entertainment.
Local Multiplayer	Two players on same keyboard. Player 1 uses arrow keys (cyan slime), Player 2 uses WASD keys (red slime). Perfect for competitive play between friends.

6. Functional Requirements

6.1 Game Field & Visual Design

Element	Specification
Field Background	Blue playing surface (#3B82F6 or similar)
Ground	Gray platform at bottom (#6B7280)
Player 1 (Left)	Cyan semicircle slime (#06B6D4)
Player 2 / AI (Right)	Red semicircle slime (#EF4444)
Ball	White or yellow circular ball with visible rotation
Goals	Net structures on both left and right edges
Visual Style	Retro arcade aesthetic with clean lines and smooth animations

6.2 Character Mechanics (Slimes)

Movement Properties

- **Shape:** Semicircle (flat bottom, curved top)
- **Horizontal movement:** Smooth acceleration and deceleration with appropriate maximum speed
- **Jumping:** Quick, responsive vertical movement with gravity affecting descent
- **Feel:** Bouncy and responsive, providing tactile feedback on player input
- **Boundary constraints:** Cannot cross center line; confined to own half of field

Grab Mechanic

- Player can grab the ball when in close proximity
- While grabbed, ball follows slime movement
- Rotational control allows aiming the release direction
- Release velocity based on slime movement and rotation input

6.3 Ball Physics

Property	Behavior
Gravity	Constant downward force creating natural arc trajectories
Bouncing	Elastic collision with ground, walls, ceiling, and slimes (configurable restitution)
Rotation	Visual spin reflecting direction/force of contact; affects trajectory on bounce
Collision Response	Velocity transfer from slime to ball based on contact point and slime velocity

Property	Behavior
Grab State	When grabbed, ball position locked relative to slime with rotational control for aiming

6.4 Controls

Action	Player 1 (Cyan)	Player 2 (Red)
Move Left	← (Left Arrow)	A
Move Right	→ (Right Arrow)	D
Jump	↑ (Up Arrow)	W
Grab/Release	↓ (Down Arrow)	S
Rotate Aim	← / → (while grabbing)	A / D (while grabbing)

6.5 Match System

Timer Configuration

- Selectable match durations: 1, 2, 3, 4, 5, 6, 7, or 8 minutes
- Countdown timer displayed prominently during gameplay
- Match ends when timer reaches zero

Scoring

- Goal scored when ball fully enters opponent's goal area
- Score display for both players visible throughout match
- Brief celebration/reset sequence after each goal
- Winner determined by highest score when time expires

6.6 Anti-Camping System

To maintain exciting gameplay, players who remain in their own goal area for extended periods receive penalties:

- **Detection:** Track time spent within own goal zone
- **Warning:** Visual indicator when approaching time limit (3-5 seconds)
- **Penalty options:** Forced ejection from goal area, temporary movement penalty, or automatic goal awarded to opponent
- **Reset:** Timer resets when player leaves goal area

6.7 AI Opponent (Single Player Mode)

- **Challenge level:** Competitive but beatable; provides satisfying challenge without frustration
- **Behavior:** Tracks ball position, predicts trajectories, makes strategic decisions
- **Imperfection:** Intentional delays and occasional mistakes to maintain fairness
- **Adaptability:** Responds to game state (attacking when behind, defending when ahead)

7. Menu System

7.1 Main Menu

- Game title/logo prominently displayed
- Single Player button
- Local Multiplayer button
- Instructions/How to Play option

7.2 Game Setup Screen

- Match duration selector (1-8 minutes)
- Start Game button
- Back to Main Menu option

7.3 In-Game HUD

- Score display (Player 1 - Player 2 / AI)
- Countdown timer
- Pause functionality with resume/quit options

7.4 End Game Screen

- Final score display
- Winner announcement
- Play Again button
- Return to Main Menu button

8. Technical Specifications

8.1 Platform Requirements

1. **Runtime:** HTML5 Canvas with JavaScript
2. **Browser support:** Chrome, Firefox, Safari, Edge (latest versions)
3. **Frame rate target:** 60 FPS for smooth gameplay

4. **Resolution:** Responsive canvas scaling with fixed aspect ratio

8.2 Physics Implementation

- Custom 2D physics engine or lightweight library (e.g., Matter.js)
- Collision detection: Circle-circle (ball-slime), circle-rectangle (ball-walls/ground)
- Semicircle collision for slime characters
- Configurable physics constants: gravity, friction, restitution, max velocities

8.3 Performance Considerations

- Optimized render loop with requestAnimationFrame
- Efficient collision detection (spatial partitioning if needed)
- Minimal garbage collection during gameplay
- Delta-time based updates for consistent physics

9. Quality Requirements

Category	Requirement
Responsiveness	Input-to-action latency < 16ms (single frame at 60 FPS)
Physics Consistency	Deterministic physics behavior; same inputs produce same results
Collision Accuracy	No clipping through walls; ball always detects goal entry correctly
Visual Smoothness	No stuttering or frame drops during normal gameplay
AI Fairness	Win rate for average player vs AI should be 40-60%

10. Success Metrics

1. **Player Engagement:** Average session length > 10 minutes
2. **Return Rate:** 50%+ of players return for multiple sessions
3. **Match Completion:** 90%+ of started matches played to completion
4. **Multiplayer Usage:** 30%+ of sessions include multiplayer mode
5. **Performance:** Consistent 60 FPS on target browsers

11. Future Considerations (Out of Scope)

The following features are not included in the initial release but may be considered for future versions:

- Online multiplayer matchmaking
- Multiple AI difficulty levels (Easy, Medium, Hard)

- Character customization (colors, accessories)
- Power-ups and special abilities
- Tournament mode
- Leaderboards and statistics tracking
- Sound effects and music
- Mobile touch controls

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